Remote sensing in practice: from the visible to the infrared for forest monitoring

UC ATHENS

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AgroParisTech

université PARIS-SACLAY

Outline

Why did you choose this course?

Presentation of the course
- Schedule
- Learning objectives
- My academic background
- Bibliography and references

General Introduction to Remote Sensing
- Historical background
- Principles of optical remote sensing
- Sentinel-2 multispectral sensor

Organization of the course

Schedule L= Lecture PW= Practical Work AM = 8:30 - 11:45 PM = 13:30 - 16:45

Monday	AM	Introduction to remote sensing (L)
		The basics of radiative transfer modeling (PW)
	PM	Inversion of leaf traits from satellites images (PW)
Tuesday	AM	Co-registration of satellite images (PW)
	PM	Interpolation of satellite time series (PW)
Wednesday	AM	Anomaly detection for forest monitoring (L)
		Start of the project
	PM	Group work outside class
Thursday	AM	Project in class
	PM	Group work outside class
Friday	AM	Project in class
		Uploading of your detection model on the website
	PM	Team presentation of the projects (evaluation)

Learning objectives

This course will cover theoretical concepts of **remote sensing** using **optical imagery** ranging from the visible to the short-wavelength infrared (0.4 µm–2.5 µm) with a focus on the practical **processing of satellite images** to detect **tree dieback** caused by bark beetle infestations.

By the end of the week, you will be able to:

- **Select** optical remote sensing data suitable **for a given application**,
- **Process** optical satellite images using **specialized Python libraries**,
- **Select and apply an anomaly detection method** for forest monitoring.

(a) 2018 (b) 2019 (c) 2020

Figure: RGB composition of Sentinel-2 images (ESA, Copernicus) - Val d'Ajol, France

My academic background

2016-20 **Engineering Degree and Research Master's**
Arts et Métiers ParisTech & Pierre and Marie Curie University

2020-23 **PhD in computer science** on the automatic mapping of impervious surfaces in cities from airborne hyperspectral images
ONERA, Magellium, Institut de Mathématiques de Toulouse, ISAE-SUPAERO

2024-25 **Postdoctoral research in computer science** on the analysis of multi-source satellite images for wildfire prediction
CNES, Laboratoire Sciences pour l'Environnement - Université de Corse - CNRS

2025- **Assistant Professor (Maître de Conférences) in computer science**
Analysis of multi-source satellite data for the monitoring of industrial pollution and agricultural crops
AgroParisTech

Bibliography and references used for this lecture

- Lecture (in French) "Le rayonnement, sa mesure et son rôle - Modélisation du Transfert Radiatif", J.P. Gastellu-Etchegorry, https://www.cesbio.cnrs.fr/dart/documentation/,
- Lecture (in French) "Télédétection et géophysique spatiale", Stéphane Jacquemoud, http://step.ipgp.fr/images/2/20/Cours.pdf,
- Lecture on Atmospheric Transmission, Simon A. Carn, https://pages.mtu.edu/~scarn/teaching/GE4250/transmission_lecture.pdf,
- CESBIO's blog: https://www.cesbio.cnrs.fr/multitemp/

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Early developments during the two world wars[1]

- **First World War** The beginnings of airborne remote sensing: ordinary cameras are mounted on reconnaissance aircraft.
- **Second World War**
  - — Mapping of coastal areas in preparation for the D-Day landings (water depth, enemy positions, camouflage nets),
  - — The German V2 surface-to-surface ballistic missile contains all the components of future space launchers.

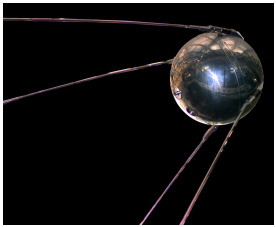
Figure: US Air Force reconnaissance aircraft during the First World War. Credit: US Air Force.

Figure: The V2 missile assembly line in the Mittelwerk complex at the Dora concentration camp. Credit: User Ericd on en.wikipedia — US Army signal corps.

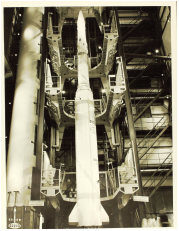
[1] https://www.esa.int/SPECIALS/Eduspace_FR, https://fr.wikipedia.org/wiki/Programme_spatial_fran%C3%A7ais

The beginning of the space race[2]

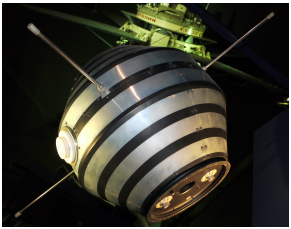
- **1957** Launch of the first artificial Earth satellite, Sputnik 1, by the USSR,
- **1958** Creation of the National Aeronautics and Space Administration (NASA) at the initiative of President Eisenhower, and launch of the first American satellite, Explorer 1.
- **1962** Creation of the Centre National d'Etudes Spatiales (CNES, French National Centre for Space Studies) at the initiative of General de Gaulle,
- **1965** Launch of the first French satellite, Astérix, with the Diamant A rocket. France becomes the third power to achieve an orbital launch with a national launcher.



(a) Replica of Sputnik 1, National Air and Space Museum.



(b) Diamant A rocket in its assembly building.



(c) Replica of Astérix, Musée de l'Air et de l'Espace, Paris Le Bourget.

[2] https://fr.wikipedia.org\/wiki\/Satellite_artificiel

- **1960** Launch of the first weather satellite TIROS-1 by NASA,
- **1964** Launch of the Meteor-1 weather satellite by the USSR,
- **1977** Launch of the European weather satellite Meteosat-1 by ESA (European Space Agency).

Figure: FCI imager radiometer of the European geostationary weather satellite MTG (third-generation Meteosat). Credit: ESA.

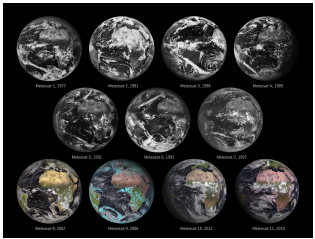


Figure: Color composites of images from the first- and second-generation Meteosat satellites. Credit: ESA/Eumetsat.

First civilian satellites for observing land surfaces

- **1972** Launch of the first land observation satellite, Landsat, by NASA, at the instigation of the United States Geological Survey and the Department of Agriculture,
- **1986** Launch of the competing SPOT satellite by CNES. In 2021, CNES released an archive of 19 million SPOT images acquired between 1986 and 2015.

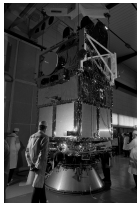


Figure: Presentation of SPOT-1 at the Toulouse Space Centre on 11 March 1985. Credit: André Cros.

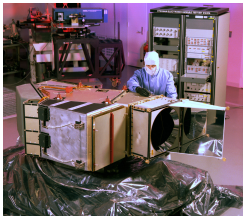


Figure: The ETM+ instrument of Landsat 7. Credit: NASA.

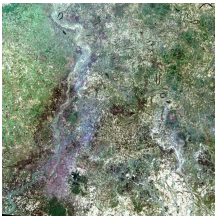


Figure: Color composite of a Landsat 7 image of Kolkata (India). Credit: NASA.

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Classification of the different types of remote sensing

- **Passive solar remote sensing**: measurement of solar radiation reflected by the Earth, *e.g.* from visible to shortwave infrared (0.4 µm - 2.5 µm),
- **Passive thermal remote sensing**: measurement of thermal radiation emitted by the Earth, *e.g.* in the thermal infrared (3 - 15 µm), or in the microwaves (220 MHz - 40 GHz).
- **Active remote sensing**: emission of lidar (visible, infrared) or radar (microwave) radiation and measurement of the backscattered radiation (*e.g.* altimeter, SAR).

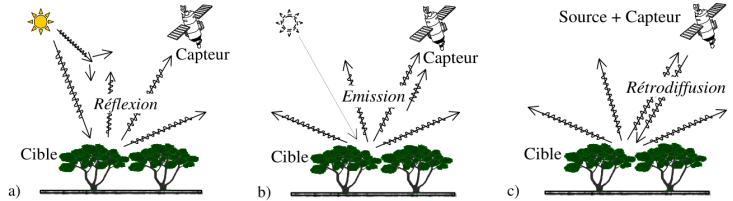


Figure: a) Passive solar, b) passive thermal, c) active. Credit: J.P. Gastellu-Etchegorry

Solar radiance

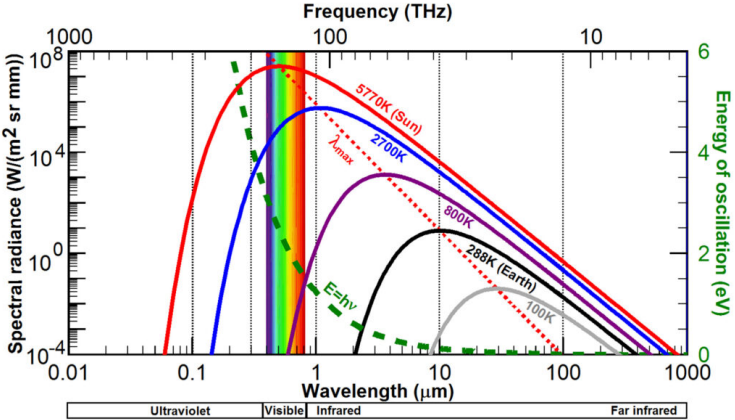


Figure: Solid lines show the radiation from a black body according to Planck's law. The red line shows the radiation for the Sun at 5770K; the black line for the Earth at 288K. The dashed red line connects the peak spectral radiances (λ_{max}) for each temperature, a consequence of Wien's displacement law. The dashed green line shows the energy ($E = h\nu$) at each wavelength according to Planck's postulate. Source: ozonedepletiontheory.info/ImagePages/electromagnetic-spectrum/

Radiance (Luminance in French) is the spectral flux that is emitted or that reaches a surface, per unit of surface, per unit of solid angle, and per unit of wavelength ($W/m^2/sr/\mu m$).

The total flux emitted by the sun is the radiance integrated over the sun surface, over the entire sphere, and over the entire electromagnetic spectrum.

Irradiance (Eclairement in French) is the flux that reaches a surface, per surface unit and per wavelength unit ($W/m^2/\mu m$).

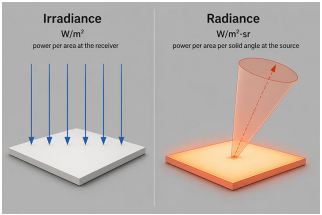


Figure: Illustration of radiance VS irradiance. Source: https://www.reddotled.com/what-is-the-irradiance-of-light.html

▸ *What is the solar irradiance at the top of the atmosphere for a surface orthogonal to the sun direction?*

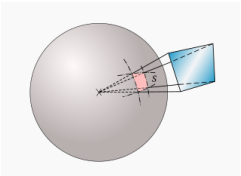
Solid angle

The solid angle is a generalization of the notion of angle in a 3D space. Given a point of view and a surface, it is defined as the ratio of the area of the surface projected on a sphere to the square of the sphere's radius:

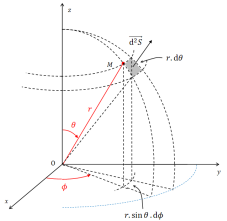
$$\Omega = S/r^2$$

The solid angle is expressed in steradians (sr), which are $\mathrm{m}^2/\mathrm{m}^2$. An elementary solid angle in spherical coordinates is defined as:

$$d^2\Omega = \frac{d^2S}{r^2} = \sin\,\theta\,\mathrm{d}\phi\,\mathrm{d}\theta$$



(a) Surface bounded by the projection of a planar quadrilateral onto a sphere, defining the solid angle. Copyright © Christophe Dang Ngoc Chan



(b) Illustration of an elementary surface in spherical coordinates. Source: https://fr.wikipedia.org/wiki/Angle_solide

Atmosphere transmittance: scattering

Atmosphere only transmits a fraction of solar irradiance because of absorption and scattering. There are mainly two types of scattering:

- **Rayleigh** scattering for molecules, proportional to λ^{-4},
- **Mie** scattering for bigger particles (*i.e.* aerosols such as dust or cloud droplets): stronger in the forward direction.

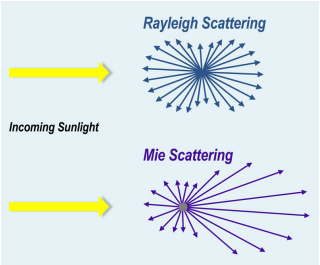
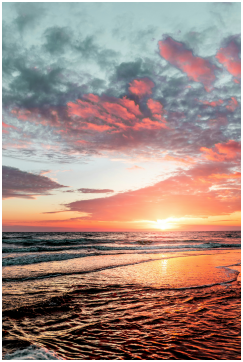


Figure: Illustration of Rayleigh and Mie scattering. Source: https://pace.oceansciences.org/prop_of_light.cgi

Atmosphere transmittance: scattering

- *What types of scattering are involved in these images?*



[2] https://unsplash.com/fr/photos/nuages-blancs-et-ciel-bleu-a33De5Sn8xo
[2] https://unsplash.com/fr/photos/plan-deau-sous-ciel-nuageux-au-coucher-du-soleil-HRX5WXFyB64

Atmosphere transmittance: absorption

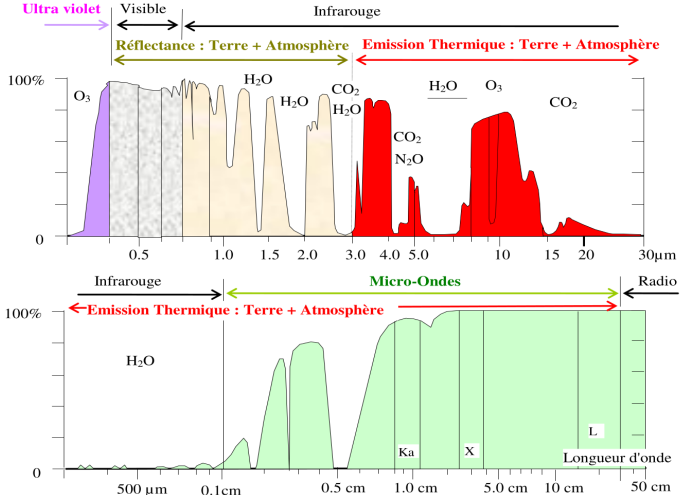


Figure: Atmospheric transmittance. Credit: J.P. Gastellu-Etchegorry

The total transmittance is the product of the transmittance due to scattering with all transmittances due to gas absorption.

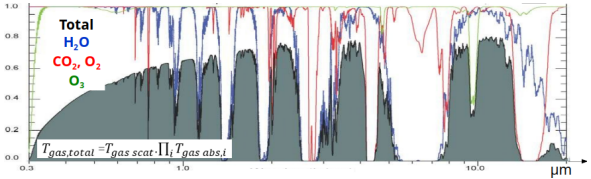


Figure: Illustration of total atmospheric transmittance (US STD76 gas model). Credit: J.P. Gastellu-Etchegorry

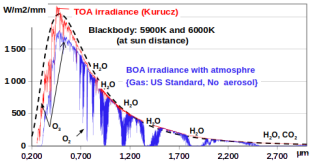


Figure: Illustration of TOA and BOA sun irradiance. Credit: J.P. Gastellu-Etchegorry

Reflectance is the physical quantity of interest: it only depends on the **surface chemical composition and rugosity**. For a surface Σ and a direction Ω, reflectance is defined as follows:

$$\rho_{\Sigma,\lambda}(\Omega) = \frac{\pi \, R_{\Sigma,\lambda}(\Omega)}{I_{\Sigma,\lambda}}$$

where $R_{\Sigma,\lambda}$ and $I_{\Sigma,\lambda}$ denote the surface radiance and irradiance at wavelength λ, respectively. We usually assume Lambertian surfaces for which reflectance does not depend on the reflected direction Ω.

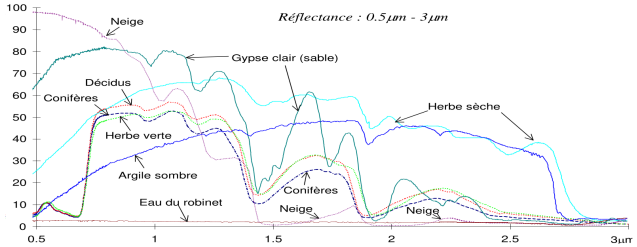
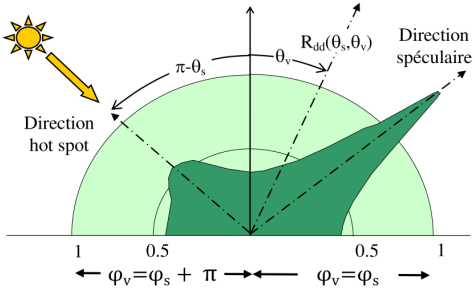


Figure: Reflectance of several materials. Credit: J.P. Gastellu-Etchegorry

Bi-directional reflectance

Natural surfaces may exhibit "hot-spot" effects or "specular" reflection.



(a) Illustration of bi-directional reflectance for a Lambertian surface and a natural surface with specular reflection. Credit: J.P. Gastellu-Etchegorry

(b) Reflectances greater than one observed over roof windows in a VENμS (CNES, IAI ELOp) image. Source: https://www.cesbio.cnrs.fr/multitemp/16885-2/

- *Show that, for a Lambertian surface, reflectance is lower or equal to one.*

As much as reflectance is intrinsic to the surface properties, it is the natural quantity to monitor the ecosystems through time, with changes in sun zenith angle, Earth-sun distance, and atmospheric variations...

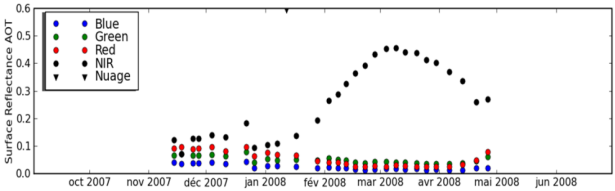
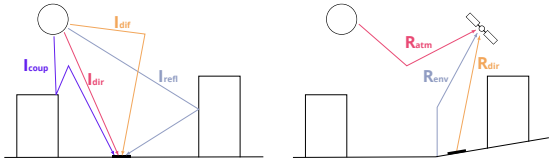


Figure: Reflectance of a wheat field as a function of time for 4 spectral bands from a time series of Formosat-2 images. Source:
https://www.cesbio.cnrs.fr/multitemp/radiometric-quantities-irradiance-radiance-reflectance/

What does the sensor measure?

Surface radiance $R_{\Sigma,\lambda}(\Omega) = \rho_{\Sigma,\lambda}(\Omega) \times I_{\Sigma,\lambda}/\pi$ is not what is measured by the sensor.



(a) Irradiance terms (b) Radiance terms

Figure: Illustration of the contributions of atmosphere to the measured signal

We have that:

$$I_{\Sigma,\lambda} = \overbrace{I_{\text{dir},\lambda}}^{I_{\text{sun,BOA},\lambda} \cos\theta_{\text{sun},\Sigma}} + I_{\text{dif},\lambda} + I_{\text{refl},\lambda} + I_{\text{coupl},\lambda}$$

$$R_{\text{sensor},\lambda} = R_{\text{atm},\lambda} + \underbrace{R_{\text{dir},\lambda}}_{R_{\Sigma,\lambda} \times \tau_{\text{dir},\lambda}} + R_{\text{env},\lambda}$$

where $\cos\theta_{\text{sun},\Sigma}$ is the sun zenith angle, and $\tau_{\text{dir},\lambda}$ is the direct upward transmittance.

Atmospheric correction codes aim to estimate the **surface-level reflectance** from the **sensor-level radiance** and the atmosphere parameters. Let us denote $I_0 = I_{\text{dir}} + I_{\text{dif}}$ the irradiance term without interactions with the Earth. Neglecting the term I_{refl}, the total irradiance can be modeled as:

$$I_\Sigma = I_0 + I_0\,\rho_{\text{env}}\,S + I_0\,(\rho_{\text{env}}\,S)^2 + \ldots = \frac{I_0}{1-\rho_{\text{env}}\,S} \quad (\text{for } \rho_{\text{env}} S < 1)$$

where ρ_{env} and S are the environment reflectance and the atmospheric backscattering coefficient also called spherical albedo (fraction of upward flux that is scattered back towards Earth). Therefore, we have that

$$R_{\text{sensor}} = R_{\text{atm}} + \underbrace{\tau_{\text{dir}} \frac{I_0}{\pi(1-\rho_{\text{env}} S)} \rho_\Sigma}_{=R_{\text{dir}}} + \underbrace{\tau_{\text{dif}} \frac{I_0}{\pi(1-\rho_{\text{env}} S)} \rho_{\text{env}}}_{=R_{\text{env}}}$$

- *Assuming that $\rho_{env} = \rho_\Sigma$, write ρ_Σ as a function of the radiative terms.*

This is usually the first step of operational atmospheric correction codes such as COCHISE[3].

[3] Poutier, L., Miesch, C., Lenot, X., Achard, V., & Boucher, Y. (2002, March). COMANCHE and COCHISE: two reciprocal atmospheric codes for hyperspectral remote sensing. In 2002 AVIRIS Earth Science and Applications Workshop Proceedings (pp. 1059-0889). Pasadena, CA, USA: Jet Propulsion Laboratory. https://popo.jpl.nasa.gov/pub/docs/workshops/02_docs/2002_Poutier.pdf

Atmospheric correction codes, *e.g.* MAJA (CNES / CESBIO), correct the measured signal from the effects of atmosphere.

(a) L1C level: Top-of-Atmosphere (TOA) reflectance data

(b) L2A level: Bottom-of-Atmosphere (BOA) reflectance data

Figure: Sentinel-2 images over Provence-Alpes-Côte-d'Azur. Credit: Copernicus, ESA, CNES.

Limitations of atmospheric correction

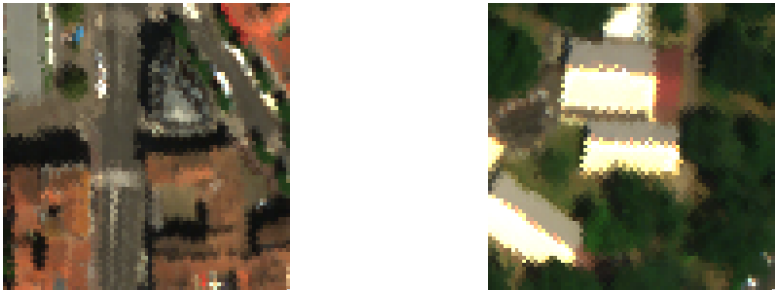


Figure: RGB composition of a hyperspectral image with a 1m spatial resolution over Toulouse, France. Credit: ONERA-DOTA, Toulouse

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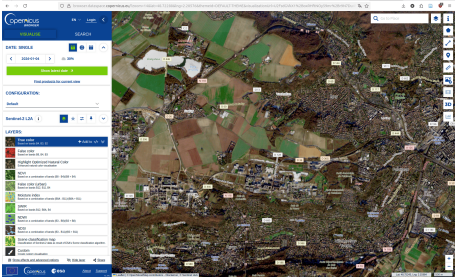
Earth observation from space: the Copernicus program

(a) https://www.copernicus.eu/fr



(b) https://browser.dataspace.copernicus.eu

Two satellites (currently three) are on a sun-synchronous orbit, allowing a global coverage at spatial resolution from 10m to 60m, and a revisit time of 5 days at the equator.

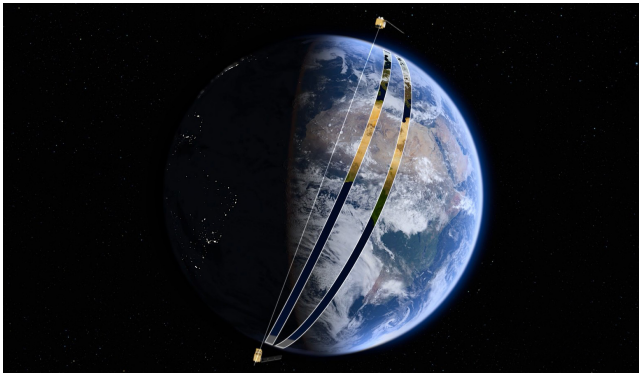


Figure: "The Sentinel-2 mission is based on a constellation of two satellites orbiting 180° apart, which along with their 290 km-wide swaths, allows Earth's main land surfaces, large islands, inland and coastal waters to be covered every five days." Credit: ESA/ATG medialab

Spectral and temporal resolution of Sentinel-2

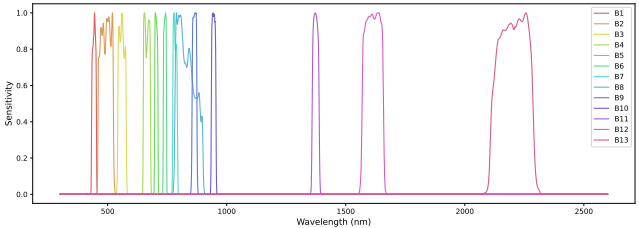


Figure: Spectral sensitivity of the Sentinel-2 sensor. The multispectral instrument has 13 bands (or spectral channels) from the visible to the short-wave infrared (SWIR).

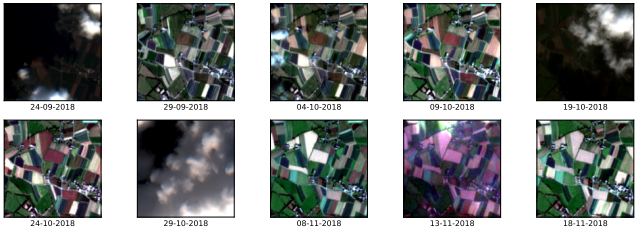


Figure: RGB composition of a Sentinel-2 (ESA, Copernicus) time series.

Let's start coding!